
VANGUARD

Autonomous Smart Perimeter Monitoring & Warning Platform

ATmega32-Based Embedded Systems Design

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Project Proposal

1. Motivation

Problem Description & Importance

Problem Description

- Conventional perimeter systems often rely on a single trigger source such as PIR or optical tripwire sensing.
- False alarms can occur because of wind, foliage, small animals, and environmental disturbances.
- Passive systems cannot aim toward the triggered sector or locally verify whether a warm object is present.

Why It Matters

- Remote sites, industrial zones, and restricted areas need fast local detection without depending fully on human response time.
- Combining motion, distance, heat, and badge authentication reduces false alarms.
- A staged verification pipeline is safer than directly acting on a single sensor trigger.

1. Motivation

Why This Project & Main Objectives

The project integrates static-sector sensing, hardware-PWM servo positioning, multi-bus communication, and interrupt-driven control on the ATmega32. The aim is a complete embedded prototype, not an isolated sensor demo.

1. **Static-sector PIR detection:** three fixed PIR sensors (INT0/INT1/INT2) each watch a static $\approx 60^\circ$ sector across the 180° field; the turret does not mechanically sweep, so a moving platform never crosses its own static thermal background.
2. **Precision lock-on:** a sector trigger slews the pan servo to that sector's center (worst case 60° from the centered idle position), then a $\pm 10\text{--}15^\circ$ horizontal micro-scan and vertical elevation micro-scan run concurrently on independent PWM channels (OC1A/OC1B) to center the narrow VL53L0X beam on the target within ≈ 650 ms.
3. **Multi-modal verification:** thermal ΔT using MLX90614 and distance R using VL53L0X over I²C.
4. **Wireless IFF handshake:** application-layer RF badge challenge-response over SPI using nRF24L01+, with a rolling counter/nonce in each payload to reject replayed transmissions.
5. **Warning actuator:** MOSFET-driven strobe/alarm output after failed authorization and positive sensor verification.
6. **Stretch features:** UART monitoring dashboard and optional MicroSD incident logging.

1. Motivation

Real-World Deployment Scenarios

Operational Sector	Threat Profile	After Sector PIR Trigger + Slew
Remote substation / solar farm perimeter	Night intrusion, cable theft, wildlife false alarms	VERIFY confirms human presence, silently reverts to SECTOR WATCH on animals/wind → IFF fails (unbadged) → strobe + incident log
Restricted depot (munitions, server vault)	Unauthorized entry, tailgating behind a badged employee	LOCK-ON micro-scan counts ToF clusters → slotted IFF window → badges < targets → alarm + escort-breach log
Automated hazard cell (robotic/welding bay)	Personnel entering an active machinery envelope	LOCK-ON + VERIFY confirm presence → IFF fails (no badges provisioned for an active zone) → interlock trip

All three share the same unmodified FSM (Section 4: Role of the ATmega32) – only badge provisioning and operator response differ by deployment.

- **Remote infrastructure watch:** multi-modal VERIFY (heat + range) filters out the wildlife/foilage false alarms that dominate single-sensor tripwires, without needing a manned response for every trigger.
- **Escort-aware access control:** the same micro-scan used for azimuth refinement doubles as a target counter, catching a side-by-side companion that a badge-only IFF check would miss; a target fully hidden directly behind another remains a known limitation of a single-beam ToF scan.
- **Industrial safety interlock:** no badges are provisioned for an active hazard zone, so any VERIFY-confirmed presence reaches WARNING OUTPUT through the same FSM; the MOSFET drives a safety relay directly, with no PC/network round-trip in the path.

2. Equipment

Hardware Components & Justification

Purpose	Exact Module Used	Justification
System Controller	ATmega32-16PU	On-chip TWI, SPI, USART, timers, interrupts, and GPIO support the complete prototype without external bus bridges.
Motion Detection	3 × PIR Motion Sensor (fixed, sector-mounted)	Fixed at $\approx 30^\circ/90^\circ/150^\circ$ sector centers, not on the moving head, so panning never sweeps the sensor's FOV across its own thermal background; edges on INT0/INT1/INT2 each command a slew to that sector.
Thermal Verification	MLX90614 IR Sensor	Factory-calibrated non-contact temperature reading over I^2C ; useful for warm-object verification.
Distance Measurement	VL53L0X Laser ToF	Digital laser time-of-flight ranging over I^2C ; less affected by air temperature and wind than ultrasonic ranging; driven with a minimal bare-metal register driver rather than the full ST API to conserve Flash.
Wireless IFF Link	nRF24L01+ Pair	Low-latency 2.4 GHz packet radio over SPI for challenge-response badge authentication.
Black-Box Logging	MicroSD Card Module	Optional high-capacity non-volatile storage over SPI for incident event logs, written via Petit FatFs / raw 512 B block writes rather than full FatFs to minimize SRAM use.
Pan-Tilt Actuation	MG996R Servos (× 2)	Standard hobby servo interface; driven by Timer1 50 Hz PWM outputs OC1A and OC1B.
Telemetry Bridge	FTDI FT232RL	Bridges ATmega32 USART to a PC USB port for the optional monitoring dashboard.
Warning Output	MOSFET + Strobe/Alarm	Low-side switch driven by GPIO to safely control a higher-current warning actuator, with a flyback diode across the inductive load to clamp switch-off voltage spikes.
Power Regulation	Dual LM2596 Converters	Two dedicated channels: one for the 5 V logic rail, one strictly for 6 V servo power, plus a Schottky diode isolating the logic rail from servo-switching transients; pan/tilt slews are staggered (never simultaneous max-torque) to stay under the servo channel's 3.5 A rating despite a ≈ 5 A combined MG996R stall current, backed by a 1000 μF bulk cap.
Level Compatibility	3.3 V Regulator / Level Shifter	Protects 3.3 V modules such as VL53L0X and nRF24L01+ when used with a 5 V ATmega32 system.

2. Equipment

Comparison with Simpler Alternatives

Subsystem	Selected Module	Simpler Alternative	Why We Chose Selected Over Alternative
Distance	VL53L0X Laser ToF	HC-SR04 Ultrasonic	Laser ToF gives compact digital ranging and is less affected by air temperature/wind than sound-based ranging.
Thermal	MLX90614 IR Sensor	LM35 Temperature Sensor	LM35 requires contact; MLX90614 supports non-contact warm-target verification.
Wireless	nRF24L01+ RF	HC-05 Bluetooth Module	nRF24L01+ supports fast packet-based custom challenge-response logic without Bluetooth pairing delay.
Logging	MicroSD Card	SPI EEPROM	MicroSD provides much larger storage for optional incident logs.
Controller	ATmega32-16PU	Arduino Uno Board	ATmega32 gives direct DIP-40 pin access and encourages register-level peripheral design.

2. Equipment

Software & Development Tools

Toolchain

- **Microchip Studio / AVR-GCC** – C development, register-level drivers, compilation
- **USBasp / eXtreme Burner** – firmware flashing
- **Proteus VSM** – circuit simulation before fabrication

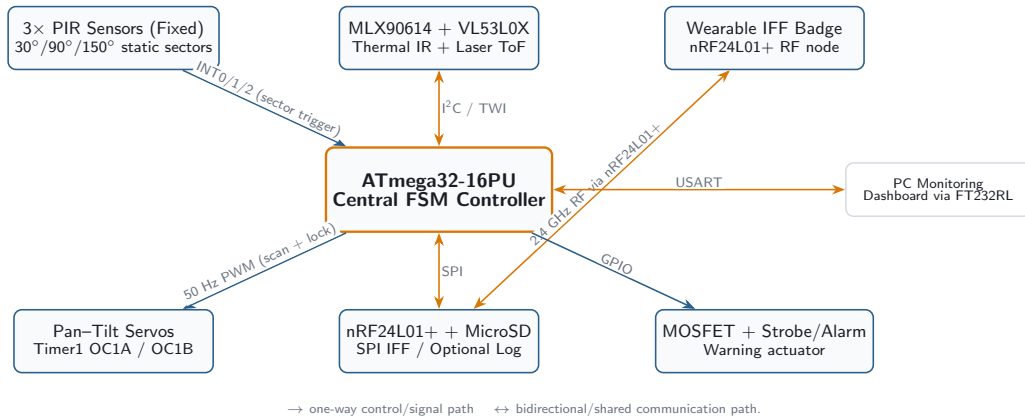
The ATmega32 performs sensing, verification, aiming, and warning control autonomously. The PC dashboard is only a visualization and debugging tool.

Host-Side Tools

- **Python 3 (PySerial)** – UART telemetry parsing
- **Pygame / Tkinter** – radar-style monitoring dashboard

3. System Design

System Block Diagram



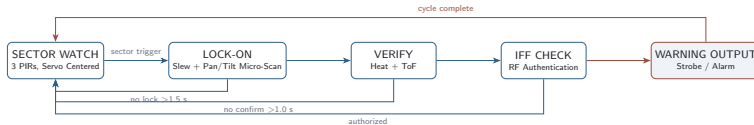
3. System Design

ATmega32 Interface Pinout Allocation Map

Pin(s)	Function	Connected To
PD0 (RXD), PD1 (TXD)	Hardware USART	FTDI FT232RL, 115200 baud
PD2 (INT0)	External interrupt 0	Fixed sector PIR 1 ($\approx 30^\circ$); commands servo slew to sector center
PD3 (INT1)	External interrupt 1	Fixed sector PIR 2 ($\approx 90^\circ$); commands servo slew to sector center
PB2 (INT2)	External interrupt 2	Fixed sector PIR 3 ($\approx 150^\circ$); commands servo slew to sector center
PD5 (OC1A)	Timer1 hardware PWM	Pan servo control signal
PD4 (OC1B)	Timer1 hardware PWM	Tilt servo control signal
PC0 (SCL), PC1 (SDA)	Hardware TWI (I ² C)	MLX90614 and VL53L0X, 7-bit addresses 0x5A and 0x29
PB5/PB6/PB7	Hardware SPI	MOSI / MISO / SCK for nRF24L01+ and MicroSD
PB4 (SS)	SPI master safety pin	Configure as output and keep HIGH
PA1	GPIO output	MicroSD chip select (CS_SD)
PA2	GPIO output	nRF24L01+ CSN
PA3	GPIO output	nRF24L01+ CE
PA4	GPIO input, optional	nRF24L01+ IRQ
PD6	GPIO output	Warning strobe/alarm MOSFET gate

4. Role of the ATmega32

Finite State Machine Logic



- Three fixed PIR sensors (INT0/INT1/INT2) each watch a static $\approx 60^\circ$ sector; the turret stays centered and does not mechanically sweep, so it never crosses its own thermal background.
- On a sector trigger, the pan servo slews to that sector's center (worst case 60° , ≈ 250 ms at the MG996R's loaded ≈ 0.2 s/ 60° rate); **LOCK-ON** then runs the horizontal and vertical micro-scans concurrently on independent PWM channels, for ≈ 650 ms total; no return within **1.5 s** reverts to **SECTOR WATCH**.
- **VERIFY** confirms agreeing heat + range readings and counts distinct ToF depth clusters from the micro-scan; **IFF CHECK** cross-checks that count against slotted badge responses ($\text{Badge_ID} \times 4$ ms + random backoff, 40 ms RX window) to flag piggybacking, or reverts within **1.0 s**.
- Further feedback paths return to **SECTOR WATCH** once the IFF is authorized or the warning cycle completes.
- **Fail-safe degraded modes:** SCL held low > 10 ms triggers a software TWI bus-reset (toggle SCL up to 9 times); an unresponsive nRF24 radio falls back to a stricter High-Alert VERIFY (double ToF check + extended ΔT hold) before WARNING OUTPUT.

4. Role of the ATmega32

Peripheral Utilization Breakdown

- **GPIO:** warning MOSFET control, status indicators, SPI chip-select/control lines.
- **External interrupts (INT0/INT1/INT2):** low-latency edge triggers from three fixed sector PIRs; each is masked during LOCK-ON/VERIFY and re-armed only after its line returns low in SECTOR WATCH, to guard against the sensor's 2.5–5 s output hold time.
- **Timer1 (16-bit):** exact 50 Hz servo PWM using OC1A and OC1B for pan and tilt.
- **Timer0:** 1 ms system tick for non-blocking task scheduling and state-machine timing.
- **Timer2:** spare timer, reserved headroom for additional timeout generation as the FSM is extended.
- **Hardware TWI (I²C):** 100 kHz bus to MLX90614 and VL53L0X, with a software timeout guard on each transaction to prevent an indefinite lock-up if a device fails to release SCL.
- **Hardware SPI:** master mode, shared by nRF24L01+ and optional MicroSD on separate CS/CSN lines; MicroSD writes are deferred/non-blocking (Petit FatFs / raw block writes, not full FatFs) so a FAT update never delays an in-flight nRF24 IFF challenge (hard RX timeout ≤ 40 ms).
- **Hardware USART:** 115200 baud telemetry frames, e.g. \$AZIMUTH,DIST,TEMP,STATUS#.

5. Implementation Plan

Milestones

Project Update 1 (Wks 1–3)

- 1.1 Bare-metal clock, GPIO, and register setup
- 1.2 Timer1 50 Hz PWM for pan-tilt servos
- 1.3 Hardware I²C driver: MLX90614 + VL53L0X
- 1.4 INT0/INT1/INT2 handlers for fixed sector PIRs, each with debounce/re-arm guard, commanding slew-to-sector

Deliverable: turret slews to the triggered sector and sends distance/temperature over UART.

Final Demonstration (Wks 4–6)

- 2.1 SPI driver for nRF24L01+ and optional MicroSD logging
- 2.2 Application-layer IFF handshake: replay-protected challenge-response, slotted multi-badge responses, ToF-cluster anti-piggyback cross-check
- 2.3 PC monitoring dashboard over UART
- 2.4 MOSFET-driven strobe/alarm warning output
- 2.5 Power isolation, level shifting, fail-safe bus/RF fallback handling, enclosure, and stress testing

Deliverable: sector PIR detection, servo slew + micro-scan aiming, thermal/range verification, IFF, warning output, and dashboard.

5. Implementation Plan

Group Member Responsibilities

Member 1: Firmware & Control

- Register-level drivers: GPIO, Timer1 PWM, Timer0 tick, I²C, external interrupts
- Core FSM architecture, incl. fail-safe/degraded-mode recovery (TWI bus reset, RF fallback)
- Timing-loop optimization

Member 2: Hardware, Power & Actuation

- Pan-tilt turret assembly
- 5 V/6 V power distribution and decoupling
- Sensor calibration, level shifting, MOSFET warning circuit

Member 3: Wireless, Storage & GUI

- SPI management for nRF24L01+ and optional MicroSD
- IFF protocol logic: replay protection, slotted multi-badge responses
- Python monitoring dashboard

6. Use of Other Controllers

Required Auxiliary IFF Badge

Main System

The ATmega32 is the sole main controller. It handles the fixed station: sector PIR detection, servo aiming, sensor verification, RF challenge initiation, optional logging, UART dashboard, and warning output.

Auxiliary Controller – Wearable IFF Badge

- **Device:** Arduino Nano / ATtiny85, battery-powered and worn by authorized personnel.
- **Why required:** the badge is physically separate from the fixed turret station, so it needs its own small controller.
- **Task:** receives a 2.4 GHz challenge packet and returns a keyed response (with a rolling counter/nonce, so a captured transmission cannot be replayed) using its own nRF24L01+ module.
- **Interface:** wireless RF only; it performs no main-system sensing, aiming, logging, or warning-output tasks.
- **Optional demo aid:** a 0.96" I²C OLED (SSD1306) shows live handshake status (IDLE / CHALLENGE RECEIVED / ACCESS GRANTED) for spectators, at zero cost to the main ATmega32.

7. Testing & Success Criteria

Measurable Criteria & Test Methodology

Metric	Target	Test Method	Priority
Detection latency	< 100 ms	Sector PIR trigger to state change (WATCH $\rightarrow$ LOCK-ON)	Essential
Azimuth capture accuracy	$\pm 5^\circ$	Final servo angle (slew-to-sector + micro-scan) vs. true bearing	Essential
Tilt lock-on time	≤ 650 ms	Sector slew + concurrent pan/tilt micro-scan, trigger-to-lock timing	Essential
Servo PWM stability	50 Hz on OC1A/OC1B	Oscilloscope waveform check	Essential
Distance verification	± 30 mm, 10–120 cm	VL53L0X indoor matte-target bench test	Essential
Thermal verification	$\Delta T \geq 3^\circ\text{C}$ above background	Controlled warm-object test	Essential
I ² C bus stability	Clean SCL @ 100 kHz	Multi-device signal-integrity check	Essential
IFF handshake execution	≤ 40 ms	RF challenge-response RX timeout	Essential
Warning output	Correct MOSFET switching	Strobe/alarm activation test	Essential
Brownout resilience	No MCU reset	Worst-case simultaneous-stall stress test (beyond the normal staggered-slew design, for margin validation)	Essential
Multi-target detection	2 distinct ToF clusters	Two-person bench test vs. badge count	Essential
Telemetry streaming rate	≥ 10 Hz, no frame loss	PC dashboard continuous update	Optional
MicroSD black-box log	Timestamp/azimuth/heat/range	File readback, no SPI collision with nRF24	Optional

Essential features: sector PIR detection, Timer1 servo aiming, I²C heat/range verification, multi-target/anti-piggyback awareness, IFF handshake, and warning output. Optional features: MicroSD logging and PC monitoring dashboard.

8. Challenges & Future Scope

Anticipated Issues & Scaling Path

Technical & Hardware Challenges

- **SPI sharing:** nRF24L01+ needs low-latency packets while MicroSD writes take longer → separate CS lines, bus-release routines, deferred writes.
- **3.3 V modules:** VL53L0X, nRF24L01+, and MicroSD SPI lines need level shifting against the 5 V bus → level shifters or safe breakout boards.
- **Memory footprint:** 2 KB SRAM is tight for full FatFs, and the official ST VL53L0X API alone costs ≈ 10 –14 KB Flash → use Petit FatFs/raw block writes and a minimal bare-metal VL53L0X driver.
- **Servo noise:** current spikes can reset the MCU → isolated servo rail, common ground, 1000 μ F bulk decoupling, flyback diode on the warning MOSFET.
- **Sensor certainty, FOV & IR interference:** MLX90614's wider FOV (≈ 35 – 90°) vs. VL53L0X's narrow beam ($\approx 25^\circ$) can catch background heat, and direct sunlight raises both sensors' IR baseline error → heat + range + IFF must agree before VERIFY, a printed shroud over both sensors, and ambient baseline recalibration during SECTOR WATCH.

Future Scope

Near-Term (Low-Cost) Enhancements

- **Status ring:** WS2812B Neopixel ring around the VL53L0X, bit-banged from 1 GPIO, visualizes FSM state live for demos.
- **Radar UI + audio:** host-side Python/Pygame polar radar with sonar-style pings from existing UART telemetry – zero added MCU cost.

Long-Term Scaling

- **Computer vision:** ESP32-CAM co-processor for visual target classification.
- **360° coverage:** stepper/continuous-rotation mechanism with slip ring support.
- **Long-range comms:** LoRa telemetry network for multi-unit perimeter monitoring.